MIL-P-43465(GL)

17 November 1966

MILITARY SPECIFICATION

PALLETS, MATERIAL HANDLING, WOOD, DOUBLE FACED,

(SPECIAL DESIGN FOR USE WITH CONEX CONTAINERS)

1. SCOPE

1.1 Scope.- This specification covers double faced 4-way pallets for use with special shipping containers.

1.2 Classification.- Pallets covered by this specification shall be of the following sizes as specified (see 6.2).

Size 1 - 31-3/8 inches long by 44 inches wide
Size 2 - 32-1/4 inches long by 57 inches wide

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

FF-N-105 - Nails, Wire, Brads, and Staples.

MILITARY

MIL-W-6110 - Wood, Determination of Moisture Content Of.

STANDARDS

FEDERAL

Fed. Std. No. 151 - Metals; Test Methods.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

FSC 3990

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MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-731 - Quality of Wood Members For Containers and Pallets

(Copies of specifications, and standards, required by suppliers in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction sample.- When specified (see 6.2), before production is commenced, a sample of the finished material handling pallet covered by this specification shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.2. The approval of the preproduction sample authorizes the commencement of production but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production item.

3.2 Materials.- Materials shall conform to referenced specifications and requirements specified herein. Materials not specifically designated shall be of the quality normally used in the manufacture of this item.

3.2.1 Wood.- Wood shall conform to group I and II of MIL-STD-731.

3.2.1.1 Quality of wood.- Quality of wood shall be determined in accordance with applicable requirements pertaining to defects and slope of cross grain as contained in MIL-STD-731 for class 1 wood members, except as otherwise specified in 3.2.1.1.1 and 3.2.1.1.2. In addition, wood shall be free from any type of current insect infestation as indicated by signs (such as: fresh dust piles, working larvae and adult insects).

3.2.1.1.1 Deckboards.- The following defects for deckboards are acceptable and take precedence over MIL-STD-731.

(a) Wane or bark not exceeding one-third the thickness, one-sixth width, and one-third the length of the piece.
(b) One shake not exceeding in length one-third the width of the piece.
(c) Sound knots with an average diameter not to exceed one-third the width of the piece.
(d) Chipped or torn grain not over 1/16 inch deep and not exceeding an aggregate area of one-sixth the surface of the piece.

(e) One split not exceeding in length, the width of the piece.

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3.2.1.1.2 Stringers.- The following defects for stringers are acceptable.

(a) Sound and tight knots not exceeding 1/2-inch in average diameter over cutouts; 1-inch elsewhere.
(b) Wane or bark not exceeding one-fourth of stringer height, providing a nailing surface of 1-3/8 inches in width remains.
(c) Checks and shake not exceeding 3/4-inch in depth and 2 inches in length.
(d) One split not exceeding 2 inches in length.

3.2.1.2 Moisture content.- At time of fabrication at the manufacturer's plant, the average moisture content of deckboards and stringers shall not exceed 19 percent. No individual piece shall have a moisture content exceeding 22 percent. Procedure for conducting moisture content testing shall be as specified in 4.3.2.1.

3.2.2 Nails.- Nails shall conform to FF-N-105, type II, style 18, four flutes, and be of hardened (heat-treated and tempered) steel having a minimum Rockwell hardness of C42 when tested in accordance with 4.4.1. The nails shall be sufficiently ductile to insure that the finished item will withstand without fracture, bending through 90 degrees over a radius not greater than the diameter of the nail when tested as specified in 4.4.2. The supplier may furnish .110 inch diameter wire plus or minus .002 inch in lieu of .120 inch diameter wire by 2 1/2 inches minimum length and may furnish a chisel point in lieu of a diamond point provided the width does not exceed the wire diameter.

3.3 Design and construction.- Design and construction of pallets shall be in accordance with the requirements specified herein and Figure 1.

3.3.1 Tolerances.- A plus or minus tolerance of 1/4 inch will be permitted on all over-all dimensions except where otherwise specified. The deckboards and stringers of any one pallet shall be uniform in thickness and height, respectively, with a tolerance of 1/8 inch over the minimum sizes specified in 3.3.2.1 and 3.3.2.2.

3.3.2 Details of components.-

3.3.2.1 Stringers.- Each pallet shall have three stringers, identical in design. Stringers shall be 2 by 4 inch nominal size, 1-1/2 inch minimum width, surfaced or smooth sawn on both top and bottom surfaces to a uniform height of not less than 3-9/16 inches. All ends shall be square. Each outside stringer shall be surfaced or smooth sawn on the outside face in addition to the top and bottom.

3.3.2.2 Deckboards.- Deckboards shall be surfaced on the outside faces. All deckboards shall be of uniform thickness, not less than 7/16 inch thick and not less than 3-5/8 inches in width, except that the two top deck end boards shall be not less than 5-1/2 inches in width. The top end deckboards

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and the bottom end entry boards shall be free of wane on their outer edges and shall be mounted flush with the ends of the stringers. Unless otherwise specified (see 6.2), spacing between top deckboards shall be not more than 1-1/2 inches. Bottom end entry boards shall be not less than 7-1/2 inches wide.

3.3.2.2.1 Chamfering.- All upper edges of the bottom deckboards shall be chamfered except for the inside edges of butted bottom deckboards (see 3.3.2.2). Each edge shall have two chamfers. Each chamfer shall be 10 inches long, centered between the stringers, and shall be cut on a 40 to 45 degree angle to the face, to within 1/8 to 3/16 inch of the bottom surface.

3.3.3 Fabrication of pallets.-

3.3.3.1 Nailing.- Load bearing points between each deckboard and stringer shall be fastened with nails specified in 3.2.2 and all nails at each point driven in a staggered pattern. For deckboards 3-5/8 to 4-15/16 inches, 5-1/2 to 6-15/16 inches, and 7-1/2 to 8-1/2 inches wide, 2, 3, and 4 nails respectively, shall be used at each point. End nails shall be driven not less than 3/4 inch from the end edges of the deckboards. Edge nails shall be driven not less than 1 inch from the side edges of the deckboards, with the remaining nails uniformly spaced between them. The heads of all nails shall be driven beneath the surface of the board. All nails bent in driving shall be removed or broken off below the surface and replaced. Nails shall not pass through the strap slots in the stringers.

3.3.3.2 Positioning of components.- Assembled pallets shall be rectangular with right angle (90 degree) corners and with all intermediate deckboards parallel to the edges. Stringer boards shall not protrude on any side.

3.4 Marking.- When specified (see 6.2), assembled pallets shall be branded with the letters "U.S." 1-inch to 1-1/2 inches high centrally located on the outside face of one of the outside stringers. Manufacturer's identification shall be neatly stenciled on the same stringer face in letters 1/4 inch to 1/2 inch high.

3.5 Workmanship.- Pallets shall be clean and assembled properly. All exposed surfaces shall be smooth and free from slivers or damage.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction sample inspection.- When a preproduction sample is required, it shall be examined for defects in table I and dimensional requirements specified herein.

4.3 Sampling for inspection.- Sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Components and material inspection.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.3.2 Intermediate tests.-

4.3.2.1 Moisture content.- Testing for moisture content shall be performed on each sample unit in accordance with MIL-M-0116. The sample unit shall be one piece of lumber. Sample units shall be obtained from each lot of lumber for each component to be used in constructing the pallets. A minimum of three determinations shall be made on each sample unit, one near each end and one at the center, with no individual reading exceeding the specified requirements. The inspection level shall be S-1 with an acceptable quality level (AQL) or 4.0, expressed in terms of defects per hundred units.

4.3.2.2 Hardness and bending of nails.- Testing for hardness and bending of nails shall be performed on each sample unit in accordance with 4.4.1 and 4.4.2 The sample unit shall be one nail. Sample units shall be obtained from the lot of nails to be used in constructing the pallets. A minimum of three determinations for hardness shall be made on each sample unit, one near each end and one at the center, with the average reading falling within the hardness range specified in 3.2.2. One determination for bending shall be made on each sample unit and the results shall be examined for compliance with requirements specified in 3.2.2. The sample plan for these physical tests shall be in accordance FF-N-105.

4.3.3 End item inspection.- The lot size shall be all pallets of each size submitted for inspection at one time. The sample unit shall be one assembled pallet.

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4.3.3.1 Visual examination.- Examination of the end item shall be in accordance with table I. The inspection level shall be level II, with an AQL of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE I.- Classification of defects

Examine	Defect	Classification Major	Minor
Wood components	Not free from insect infestation.	X	
Deckboards	More than one shake.	X	
	Evidence of rot.	X	
	Shake exceeding in length one-third the width of the piece.	X	
	Through check exceeding in length the width of tne piece.	X	
	Wane or bark exceeds one-third the thickness, one-sixth the width and one-third the length of the piece.	X	
	More than one split or split exceeds in length the width of the piece.	X	
	Slope of grain exceeds 1 inch in 10 inches.	X	
	Unsound knot exceeding in diameter 2 inches or 1/4 the width of the piece.	X	
	Sum of widths of holes (worn holes, knotholes, unsound knots, hollow knots, or other) exceed that which is specified for knots for class 1 of MIL-STD-731.	X	
	Average diameter of sound knots exceeds 1/3 the width of the piece.	X	
	Chipped or torn grain exceeding 1/16 inch depth and exceeding an aggregate area of 1/6 the surface of the piece.	X	
	Boards not surfaced where required.	X	
	Bottom deckboards not chamfered on upper edge.	X	

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TABLE I.- Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Stringer	Sound or tight exceeding 1/2 inch in diameter over cutouts; 1-inch elsewhere.	X	
	Evidence of rot.	X	
	Slope of grain exceeds 1-inch in 10 inches.	X	
	Wane or bark exceeding 1/4 of stringer height; less than 1-3/8 inches nailing surface remaining.	X	
	Unsound knot exceeding in diameter 2 inches or 1/4 the width of the piece	X	
	Check of shake exceeding 3/4 inch in depth or 2 inches in length.	X	
	More than one split in piece, or split exceeds 2 inches in length.	X	
	Stringers not surfaced where required.	X	
Nails	Not type or size specified.	X	
	Not staggered or positioned as specified.	X	
	Not of four flutes.	X	
	Chisel point exceeds wire diameter (when applicable).	X	
Assembly	Assembled unit does not form a rectangle with 90 degree corner.	X	
	Intermediate boards not parallel to sides.		X
	Stringer protrudes on any side.		X
Workmanship Marking (and branding when applicable)	Type, size and location not as specified, not legible, or permanent, when required.		X
	Not clean or smooth.		X
	Exposed surface contains slivers or damage.	X	

4.3.3.2 Dimensional examination.- Examination shall be made of pallets for compliance with dimensions specified. Any noncompliance with specified requirements shall constitute a defect. The inspection level shall be S-2 with an AQL of 6.5, expressed in terms of defects per hundred units.

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4.3.4 Examination of preparation for delivery.- Inspection to determine compliance with preparation for delivery requirements shall be accomplisned in accordance with table II and section 5. Defects will be scored as specified in table II. The inspection level shall be S-4 with an acceptable quality level of 6.5 defects per hundred units.

TABLE II.- Defects for packing and marking

Examination	Defect
Markings	Incorrect; illegible; of improper size, location, sequence or method of application.

4.4 Tests.-

4.4.1 Hardness.- Tests for hardness shall be in accordance with the applicable requirements of method 243.1 of Federal Test Method Standard No. 151.

4.4.2 Bending.- Tests for bending shall be in accordance with the applicable requirements of method 231.1 of Federal Test Method Standard No. 151.

5. PREPARATION FOR DELIVERY

5.1 The pallets shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The method of packing shall be in accordance with rules or regulations of carriers applicable to the mode of transporation.

5.2 Marking.- In addition to any special marking required by the contract or order, shipments shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- Pallets covered by this specification are intended for use in conjunction with triple wall corrugated boxes conforming to MIL-B-40642, within metal shipping containers conforming to MIL-B-11886.

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6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Size required (see 1.2).
(c) When a preproduction sample is required (see 3.1).
(d) When spacing of top deckboard is to be different (see 3.3.2.2).
(e) When brand marking is required (see 3.4).

Custodian:

Army - GL

Preparing activity:

Army - GL

Review activities:

Army - SM, ME

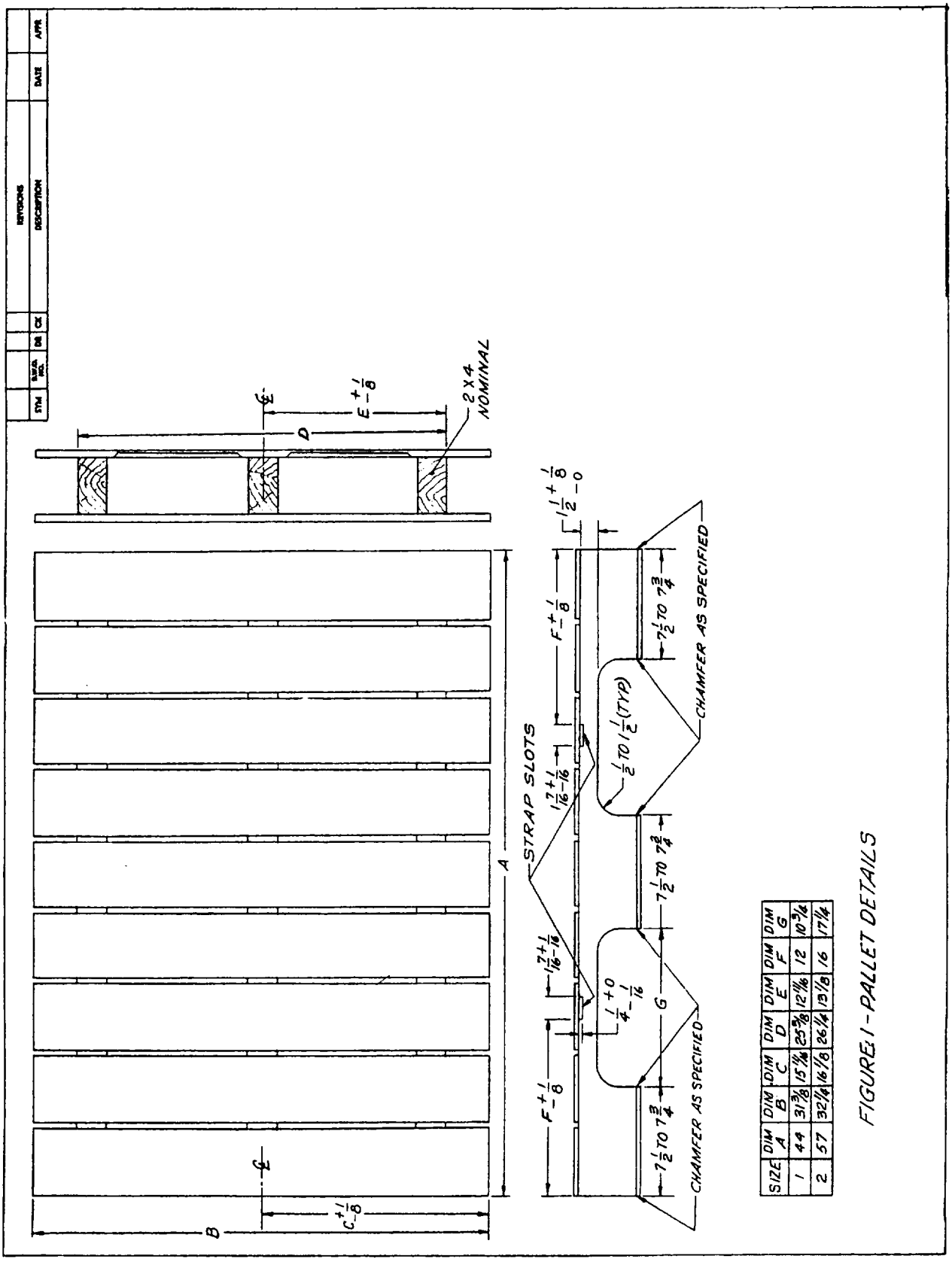
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SIZE	DIM A	DIM B	DIM C	DIM D	DIM E	DIM F	DIM G
1	44	31 3/8	15 7/8	25 3/8	12 7/8	12	10 3/4
2	57	32 1/4	16 1/8	26 1/4	13 1/8	16	17 1/4

FIGURE 1 - PALLET DETAILS

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION
Pallets, Material Handling, Wood, Double Faced, (Special Design For Use With Conex Containers) - MIL-P-43465(GL)

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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